Mohr's Salt as a Wide-Coverage Testing Material for Analytical Chemistry Courses

Mohr's salt (ammonium iron(II) sulfate hexahydrate, $Fe(NH_4)_2(SO_4)_2{\cdot}6H_2O$) is recommended as a testing material for the lab curriculum of the analytical chemistry class in college. We have students individually prepare Mohr's salt as a common testing material. Using the Mohr's salt they have prepared, the students are able to perform a wide range of lab experiments in analytical chemistry; that is, they learn volumetric, gravimetric, and instrumental analyses through the determinations of iron and sulfate in Mohr's salt.

Syllabus for a Lab Class Using Mohr's Salt

Inorganic Chemistry	
Preparation and Purification of Mohr's salt	$(NH_4)_2SO_4 + FeSO_4 + H_2O \xrightarrow[\text{heat}]{\text{Fe wire}} Fe(NH_4)_2(SO_4)_26H_2O$
	Recrystallize twice from water, air-dry on filter paper at room temp. or in oven below 50 °C.
Volumetric Analysis	
Redox Titration	Standardization of $KMnO_4$ soln. against standard sodium oxalate soln. Determination of Fe(II) in student-prepared Mohr's salt by permaganometry.
Chelatometric Titration	Oxidation pretreatment of student-prepared Mohr's salt with hydrogen peroxide (Fe(II)→Fe(III)) Titration of Fe(III) with standard EDTA soln. by using of variamine blue B as an indicator.[a]
Gravimetric Analysis	Preparing, ignition, and weighing of $BaSO_4$.
Instrumental Analysis	
Colorimetry	Measurement of visible absorption spectrum of Fe(II)–phenanthroline with standard Mohr's salt. Making a calibration curve against standard Fe(II)–phenanthroline. Determination of Fe(II) content in the student-prepared Mohr's salt.
Potentiometry	Standardization of ammonium cerium(IV) sulfate soln. with standard Mohr's salt soln. Potentiometric titration of the student-prepared Mohr's salt soln. with the standardized ammonium cerium(IV) sulfate soln. using Pt electrode.[b]

[a] Erdey, L.; Rady, G. *Z. Anal. Chem.* **1956,** *149,* 250.
[b] Laitinen H. A. *Chemical Analysis*; McGraw-Hill: New York, 1960.

This lab class may be used as a combination curriculum of inorganic and analytical chemistry by consideration of preparation and purification of Mohr's salt. This curriculum is suitable for a one-credit half-semester course of two three-hour lab classes in a week or for a one-credit one-semester course consisting of one three-hour lab class a week. An example is shown in the table.

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